

CHAPTER 7

Crystals, Lattices, Lattice Vibrations
and Phonons

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Schrödinger's Equation-Fundamentals

If a quantum system (such as an atom) is known to be in a particular quantum state s , we can describe all of the physical properties of the system in terms of the wavefunction $\Psi_s(\mathbf{r}, t)$, which obeys Schrödinger's equation

$$i\hbar \frac{\partial \psi_s(r, t)}{\partial t} = \hat{H} \psi_s(r, t),$$

$$\hat{H} = \hat{H}_0 + \hat{V},$$

Where $\hat{H}_0$ is the Hamiltonian of a free atom and V is the interaction with the external field. For the case in which no external field is applied to the atom, the Hamiltonian is simply equal to $\hat{H}_0$ and Schrödinger equation possesses solution in the form of energy eigenstates, the time of evolution of these states is given by a simple exponential phase factor

$$\psi_n(\mathbf{r}, t) = u_n(\mathbf{r}) \exp(-i\omega t),$$

By substituting this form wavefunction into the Schrödinger's equation, we find that the spatially varying part of the wavefunction $u_n(\mathbf{r})$ must satisfy the eigenvalue equation (Known as time-independent Schrödinger's equation)

$$\hat{H}_0 u_n(\mathbf{r}) = E_n u_n(\mathbf{r}),$$

Where $E_n = \hbar\omega_n$, $u_n(\mathbf{r})$ are called the energy eigensolutions and are assumed to be the orthonormal

$$\int u_m^*(\mathbf{r}) u_n(\mathbf{r}) d^3r = \delta_{mn},$$

The wavefunction of state s , $\Psi_s(\mathbf{r}, t)$, can be expressed in terms of eigenstates $u_n(\mathbf{r})$

$$\psi_s(\mathbf{r}, t) = \sum_n C_n^s(t) u_n(\mathbf{r}),$$

The expansion coefficient gives the probability amplitude that the atom is in the energy eigenstate n at time t .

7.1 Adiabatic Approximation

- Schrodinger equation to describe a semiconductor

The Hamiltonian:

$$H = -\frac{\hbar^2}{2} \sum_{j=1}^M \frac{1}{M_j} \Delta_{\mathbf{R}_j} - \frac{\hbar^2}{2m_0} \sum_{i=1}^N \Delta_{\mathbf{r}_i} + \frac{1}{4\pi\epsilon_0} \times \left(\sum_{j>j'} \frac{e^2 Z_j Z_{j'}}{|\mathbf{R}_j - \mathbf{R}_{j'}|} + \sum_{i>i'} \frac{e^2}{|\mathbf{r}_i - \mathbf{r}_{i'}|} + \sum_{i,j} \frac{e^2 Z_j}{|\mathbf{R}_j - \mathbf{r}_i|} \right) \quad (7.1)$$

So: $H\phi(\mathbf{r}_i, \mathbf{R}_j) = E\phi(\mathbf{r}_i, \mathbf{R}_j)$ (7.2)

To solve this equation, we must use some approximations:

(1) Adiabatic or Born-Oppenheimer approximation:

$$M_j ; 1836 \cdot A_j m_0 \quad (7.3)$$

The resonance frequencies of ions oscillate are much lower than electrons:

$$\Omega \simeq (\beta M_j^{-1})^{\frac{1}{2}} \ll \omega = (\beta m_0^{-1})^{\frac{1}{2}} \quad (7.4)$$

The electrons can practically instantaneously follow the motion of the ion cores, but not vice versa.

- Then we separate $\phi(\mathbf{r}_i, \mathbf{R}_j)$ into :

$$\phi(\mathbf{r}_i, \mathbf{R}_j) = \phi(\mathbf{r}_i) \phi(\mathbf{R}_{j0}) \quad (7.5)$$

(2) Single electron approximation

$$H_{1,e}(\mathbf{r}) = E\phi(\mathbf{r})$$

$$\left[\frac{\hat{p}^2}{2m} + V(\mathbf{r})\right]\phi(\mathbf{r}) = E\phi(\mathbf{r})$$

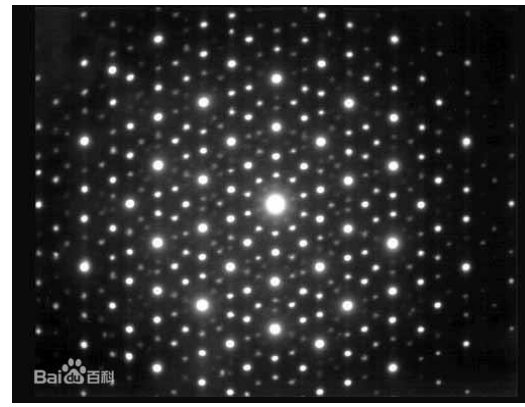
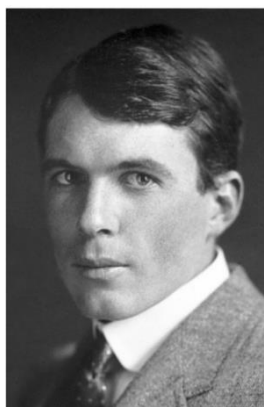
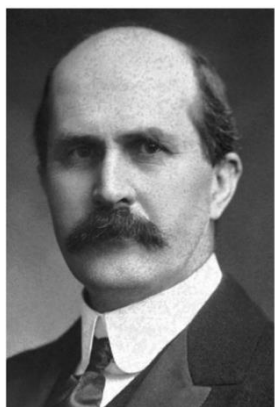
$$\hat{p} = -i\hbar\nabla$$

$$V(\mathbf{r} + \mathbf{R}) = V(\mathbf{r})$$

Crystal = Bravais Lattice + Primitive Cell

- ❑ The most fundamental characteristic of a crystal structure is the periodicity in the spatial arrangement of its atoms, ions, or molecules. The basic repeating unit within the crystal is approximated as a geometric point. The set of points obtained this way exhibits periodic repetition in three-dimensional space, thereby forming a **lattice, known as Bravis Lattice**
- ❑ A lattice extends infinitely in terms of spatial distribution, meaning the number of lattice points it contains is infinite
- ❑ Connecting any two points in a lattice forms a vector. When this vector is translated in any direction, if one end of the vector falls on any lattice point, the other end must land on another lattice point within the lattice
- ❑ Every lattice point has an identical surrounding environment
- ❑ In a crystal lattice, select a lattice point as a vertex and form a parallelepiped as a repeating unit, with the edges being the periods along three non-coplanar directions. When this parallelepiped undergoes periodic translation along the three directions, it can fill the entire lattice to form a crystal—this parallelepiped is known as the primitive cell (or "unit cell" in a broader context, though "primitive cell" specifically refers to the smallest repeating unit with only one lattice point)
- ❑ Crystal = Bravais Lattice + Primitive Cell
- ❑ In all crystal structures, there are only 230 distinct combinations of symmetry elements, known as the 230 space groups. These were independently derived by the Russian crystallographer Evgraf Fedorov and the German crystallographer Artur Moritz Schoenflies For this reason, they are also called the "230 Fedorov Groups"
- ❑ The crystal structure can be determined using X-ray diffraction

- Max von Laue discovered the phenomenon of X-ray diffraction in crystals. This discovery not only proved the wave nature of X-rays but also confirmed the lattice structure of crystals. He derived a mathematical equation known as the **Laue equations**. In 1914, von Laue was awarded the Nobel Prize in Physics for his discovery of X-ray diffraction in crystals
- At the end of 1912, William Henry Bragg and his son William Lawrence Bragg proposed a theoretical explanation for the phenomenon of X-ray crystal diffraction—the renowned **Bragg's Law**. Based on this theory, the elder Bragg built an X-ray spectrometer and began analyzing the crystal structures of inorganic substances.
- After his father's death, the younger Bragg gradually shifted his research focus to biological macromolecules. He believed X-rays held great potential for determining the structure of biological macromolecules and took the lead in establishing the Medical Research Council Laboratory of Molecular Biology (MRC LMB) at the University of Cambridge
- This laboratory attracted a young researcher interested in protein structures in 1948. Three years later, his collaborator also joined the laboratory. Their names were Francis Crick and James Watson. They discovered the double-helix structure of DNA, discovered by Francis Crick and James Watson, ushered molecular biology into a brand-new era
- Beyond this, the Molecular Biology Laboratory established under the younger Bragg's leadership has produced a total of 12 Nobel laureates. Additionally, 11 other Nobel Prize winners have worked at this laboratory at some point in their careers



7.2 Lattices and Crystal Structures in Real and Reciprocal Space

- Crystalline solids have a periodic spatial arrangement of atoms, we reach an identical atom by a vector :

$$\mathbf{R} = n_1 \mathbf{a}_1 + n_2 \mathbf{a}_2 + n_3 \mathbf{a}_3 \quad \text{with } n_i = 0, \pm 1, \pm 2, \dots \quad (7.6)$$

The vector $\mathbf{R}$ is called a translation vector of the lattice, The vectors $\mathbf{a}$, define a parallelepiped which is called the unit cell

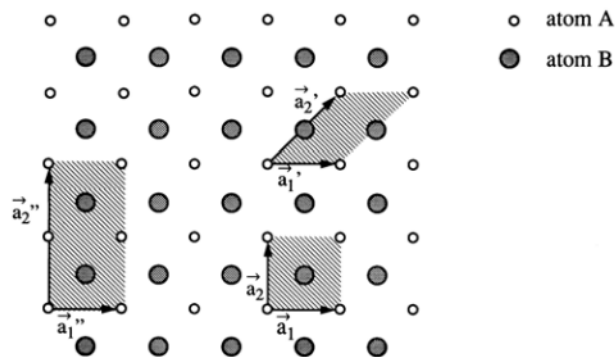


Fig. 7.1. Two primitive (*r.h.s.*) and a non-primitive (*l.h.s.*) unit cells in a two-dimensional cubic lattice with a basis consisting of two different atoms per primitive unit cell

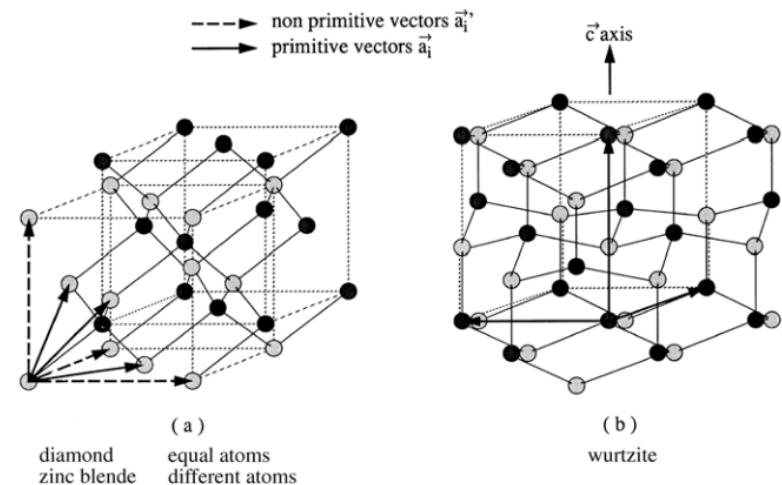


Fig. 7.2. The unit cells for the diamond and zinc-blende-type crystal structures (a) and of the wurtzite crystal structures (b) (see [82L1] of Chap. 1)

- The volume of the unit cell given by: $V_{uc} = \mathbf{a}_1 \bullet (\mathbf{a}_2 \times \mathbf{a}_3)$ (7.8)
- Translational group: The vectors $\mathbf{R}$ form, for an infinite crystal, an Abelian group
- Point group: Another type of symmetry operation, and these operations form a group

The seven crystal systems

One can divide the [Bravais lattices](#) into 7 **crystal systems**, which are defined by the lengths a , b , c and angles α , β , γ between the [primitive translation vectors](#). So one classifies different [lattices](#) according to the shape of the parallelepiped spanned by its [primitive translation vectors](#).

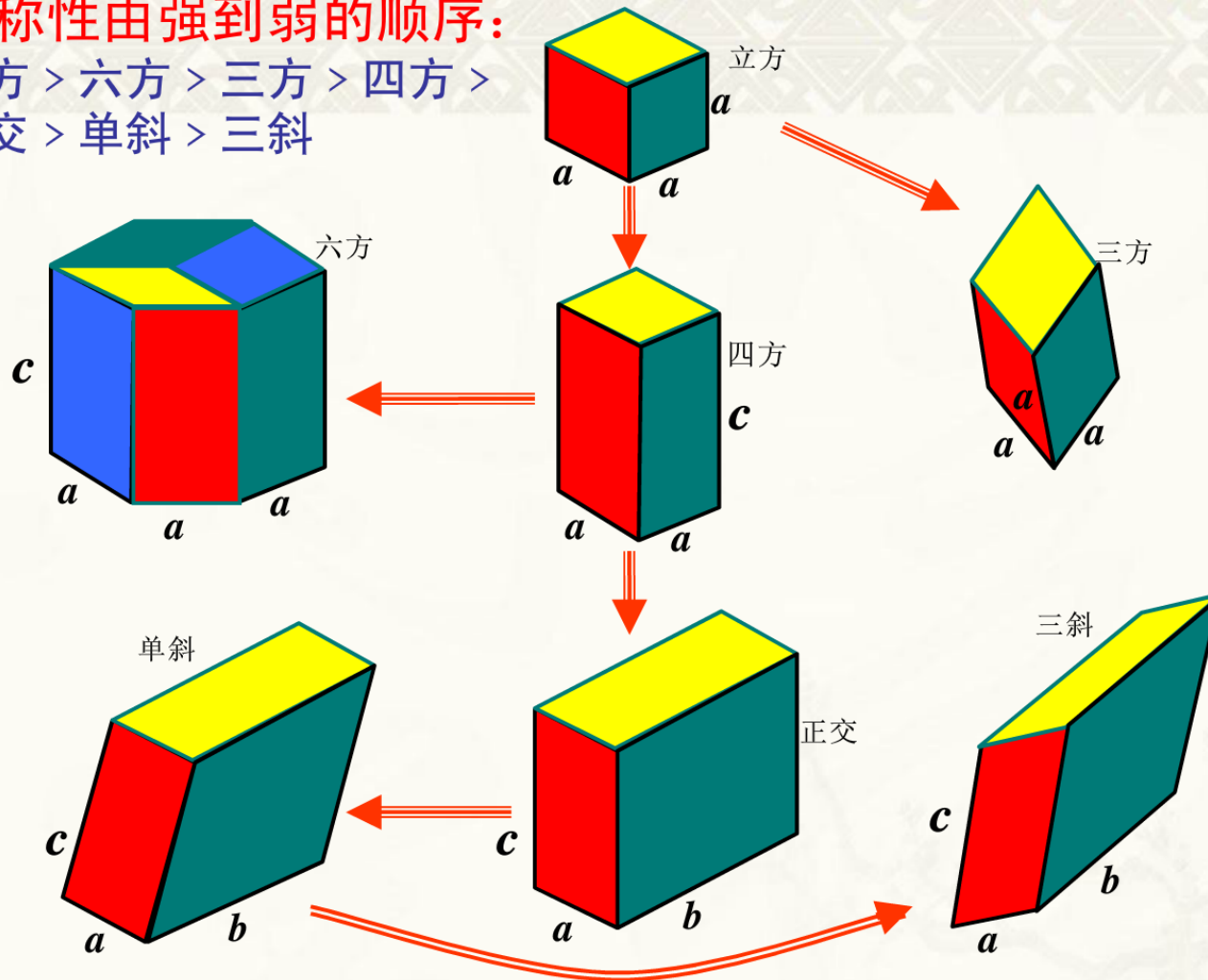
Crystal System	Lengths	Angles
Cubic (立方)	$a=b=c$	$\alpha=\beta=\gamma=90^\circ$
Trigonal (三方)	$a=b=c$	$\alpha=\beta=\gamma < 120^\circ, \neq 90^\circ$
Hexagonal (六方)	$a=b \neq c$	$\alpha=\beta=90^\circ, \gamma=120^\circ$
Tetragonal (四方)	$a=b \neq c$	$\alpha=\beta=\gamma=90^\circ$
Orthorhombic (正交)	$a \neq b \neq c$	$\alpha=\beta=\gamma=90^\circ$
Monoclinic (单斜)	$a \neq b \neq c$	$\alpha=\beta=90^\circ \neq \gamma$
Triclinic (三斜)	$a \neq b \neq c$	$\alpha \neq \beta \neq \gamma$

So sometimes it makes sense not to use a [primitive unit cell](#) but one which fits better to the symmetry of the problem. This idea leads to the **14 Bravais Lattices**.

7 crystal systems

对称性由强到弱的顺序:

立方 > 六方 > 三方 > 四方 >
正交 > 单斜 > 三斜



- Reciprocal lattice (First proposed by German physicist Paul Peter Ewald in 1913) :

$$\mathbf{b}_1 = \frac{2\pi}{V_{uc}} \mathbf{a}_2 \times \mathbf{a}_3 \quad (7.7)$$

A general translation vector of the reciprocal lattice is:

$$\mathbf{G} = l_1 \mathbf{b}_1 + l_2 \mathbf{b}_2 + l_3 \mathbf{b}_3 \quad l_i = 0, \pm 1, \pm 2, \dots \quad i = 1, 2, 3. \quad (7.9)$$

The connection between real space and reciprocal

- Because of periodic function in real space: $f(\mathbf{R} + \mathbf{r}) = f(\mathbf{r})$
can be written as a Fourier series summing over all vectors of the reciprocal lattice .

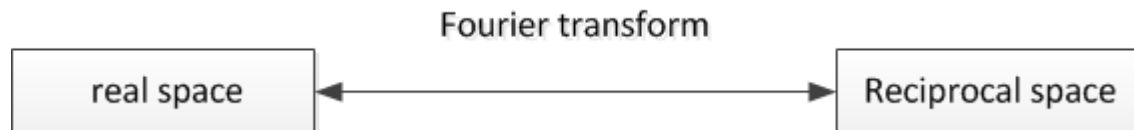
with:
$$f(\mathbf{r}) = \sum_{\mathbf{G}} f_{\mathbf{G}} e^{i\mathbf{G}\mathbf{r}} \quad (7.10)$$

and also :

$$f_{\mathbf{G}} = V_{uc}^{-1} \int_{uc} f(\mathbf{r}) e^{-i\mathbf{G}\mathbf{r}} d\tau$$

$$\mathbf{R} \cdot \mathbf{G} = 2\pi m; \quad m = 0, \pm 1, \pm 2, \dots \quad (7.11)$$

SO:
20:



7.3 Vibrations of a String

- The lattice vibrations .
- First consider a quasi one-dimensional string, transverse and longitudinal wave.

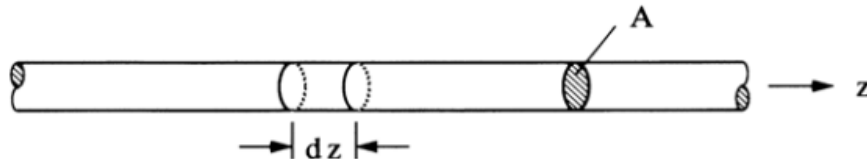


Fig. 7.5. A piece of a string as a model to explain the derivation of (7.19)

Then Newton's equation of motion reads:

$$dm \frac{\partial^2 u}{\partial t^2} = \rho A \cdot dz \cdot \frac{\partial^2 u}{\partial t^2} = F \quad (7.16)$$

The force F is connected to the elasticity modulus E via stress: $\sigma(z) = E \frac{\partial u}{\partial z}$ (7.17)

So:
$$F = A \frac{\partial \sigma}{\partial z} = A \cdot E \cdot \frac{\partial^2 u}{\partial z^2}$$

$$\rho \frac{\partial^2 u}{\partial t^2} = E \frac{\partial^2 u}{\partial z^2} \quad (7.19)$$

For a plane wave, $u = u_0 \exp[i(kz - \omega t)]$, the dispersion relation for longitudinal wave:

$$\omega_L = (E/\rho)^{1/2} k \quad (7.21)$$

- The phase and group velocity : $v_{ph}^L = v_g^L = (E/\rho)^{1/2}$ (7.22)
- For two degenerate, transverse waves :

$$\omega_T = (G/\rho)^{1/2} k \quad (7.23)$$

or $v_{ph}^T = v_g^T = (G/\rho)^{1/2}$ (7.24)

Since: $G \leq E$, we find: $v_{ph}^T \leq v_{ph}^L$ (7.26)

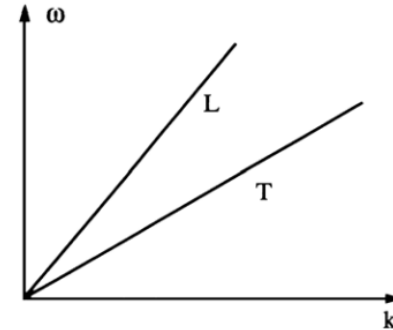


Fig. 7.6. The dispersion relation of waves on a homogeneous string

We now should consider the regime of validity of the above calculation. We assumed a homogeneous string, neglecting the fact that a solid is made up from atoms. Therefore the above approximation can only be valid for wavelengths much longer than the lattice constant or for wave vectors close to the center of the first Brillouin zone, i.e.,

$$\lambda \gg a \quad \text{or} \quad k \ll \frac{\pi}{a} . \quad (7.27)$$

7.4 Linear Chains

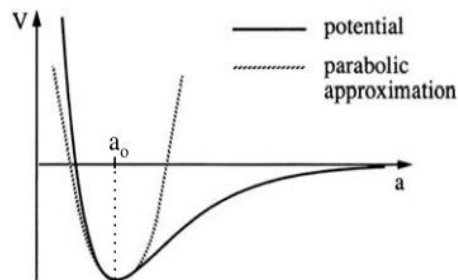


Fig. 7.7. Sketch of the potential V between neighboring atoms in a solid as a function of the lattice constant a

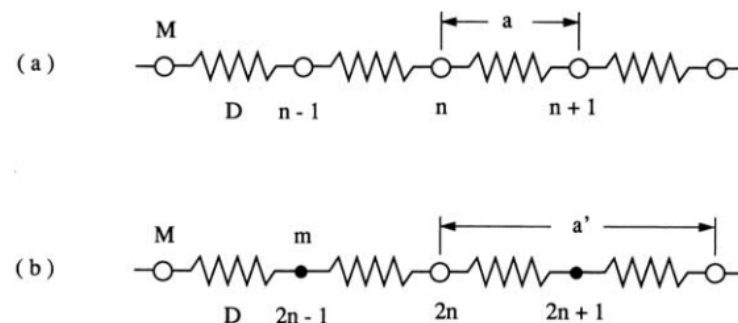


Fig. 7.8. The linear chain model for the cases of one atom per unit cell (a) and two atoms per unit cell (b)

In Fig.7.7: The equilibrium position a_0 , the state close to the energetic minimum.

In Fig.7.8: Every atom with mass M is connected to its neighbors with a force constant D .

The displacement of atom u_n from the equilibrium position :

$$M \frac{\partial^2 u_n}{\partial t^2} = D[(u_{n+1} - u_n) - (u_n - u_{n-1})] = D[u_{n+1} - 2u_n + u_{n-1}] \quad (7.28)$$

For a plane wave , the solution can be:

$$u_n = u_{n,0} \exp[i(kna - \omega t)] \quad (7.29a)$$

$$u_{n\pm 1} = u_{n\pm 1,0} \exp[i(k(n\pm 1)a - \omega t)] \quad (7.29b)$$

- For a plane wave, we conclude that: $u_{n,0} = u_{n+1,0} = u_{n-1,0}$ (7.29c)
Inserting (7.29a,b,c) into (7.28) gives:

$$-M\omega^2 = D(e^{-ika} - 2 + e^{ika}) = -2D(1 - \cos ka) \quad (7.30)$$

$$\text{or } \omega = \left(\frac{4D}{M} \right)^{1/2} \left| \sin \frac{ka}{2} \right| \quad (7.31)$$

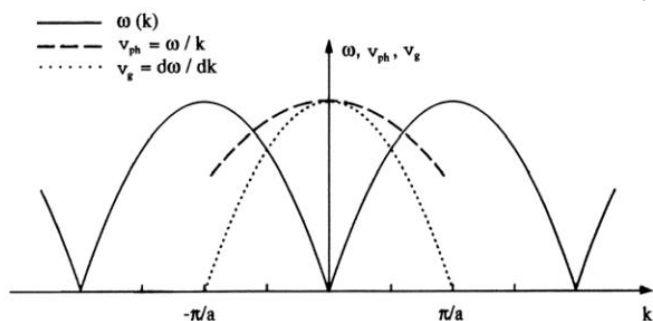


Fig. 7.9. The dispersion relation and the phase and group velocities of the longitudinal vibrations of the monatomic chain of Fig. 7.8a

- The dispersion relation outside the first Brillouin zone repeats just what is inside.
- If adjacent atoms are displaced in antiphase, we get :

$$\lambda_{\min} = 2a \rightarrow k_{\max} = \frac{2\pi}{\lambda_{\min}} = \frac{\pi}{a} \quad (7.32)$$

- The situation with more than one atom per primitive unit cell with atoms M and m. From Fig.7.8b: we obtain the equations of motion in analogy to (7.28)

(7.33a)

$$M \frac{\partial^2 u_{2n}}{\partial t^2} = D(u_{2n+1} - 2u_{2n} + u_{2n-1})$$

$$m \frac{\partial^2 u_{2n+1}}{\partial t^2} = D(u_{2n+2} - 2u_{2n+1} + u_{2n}) \quad (7.33b)$$

using again the ansatz:

$$u_{2n} = u_{2n,0} \exp[i(2nak - \omega t)] \quad (7.34a)$$

$$u_{2n+1} = u_{2n+1,0} \exp\{i[(2n+1)ak - \omega t]\} \quad (7.34b)$$

And noting that the amplitudes of equal masses are equal:

$$u_{2n,0} = u_{2n+2,0} = A_M \quad (7.35a)$$

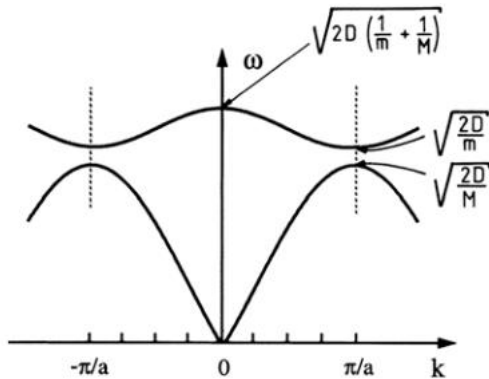
$$u_{2n+1,0} = u_{2n-1,0} = A_m \quad (7.35b)$$

We get :

$$\begin{aligned} (2D - \omega^2 M) A_M - 2D \cos(ka) A_m &= 0 \\ -2D \cos(ka) A_M + (2D - \omega^2 m) A_m &= 0 \end{aligned} \quad (7.36)$$

The dispersion relation resulting from the corresponding secular equation reads:

$$\omega^2 = D \left(\frac{1}{m} + \frac{1}{M} \right) \pm D \left[\left(\frac{1}{m} + \frac{1}{M} \right)^2 - \frac{4}{Mm} \sin^2 \frac{ka}{2} \right]^{1/2} \quad (7.37)$$



The lower branch: acoustic branch
The upper branch: optical branch

Fig. 7.11. The dispersion relation for the diatomic chain

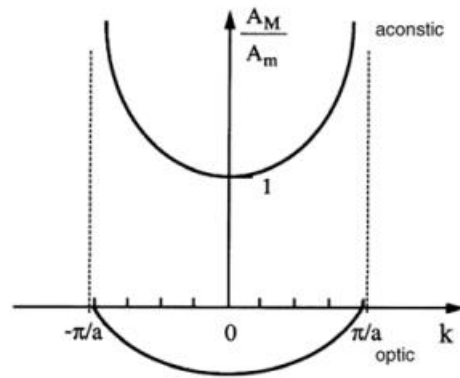


Fig. 7.12. The ratio of the amplitudes A_M/A_m of the two different masses for the two dispersion branches of Fig. 7.11 as a function of k

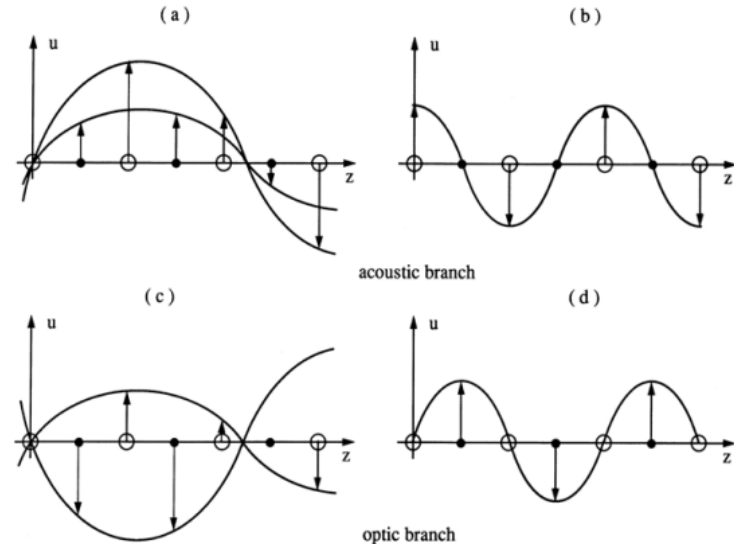


Fig. 7.13. Sketch of the displacements of the atoms on the acoustic (a, b) and the optical branches for two different wave vectors (c, d)

- Discuss the connection between 7.12 and 7.13:
 1. For the acoustic branch, the two different atoms are displaced in the same direction. For increasing k , the amplitude of M gets larger than m . At the boundary of the first Brillouin zone only the heavy masses oscillate.
 2. For the optical branch, the two atoms are displaced in anti-phase. At the boundary of the first Brillouin zone only the light masses oscillate.
- Both for the monatomic and the diatomic chain, for every wave vector and every branch, two (degenerate) transverse oscillations also exist

7.5 Three-Dimensional Crystals

- The atoms are connected with nearest (and possibly next-nearest) neighbors in the three-dimensional crystal structure.
- The result of the equations analogous to (7.28) or (7.33) will be qualitatively the same.
- Three acoustic branches:

$$\left. \begin{array}{l} \text{number of acoustic branches: } 3 \\ \text{number of optical branches: } 3s - 3 \end{array} \right\} \quad (7.39)$$

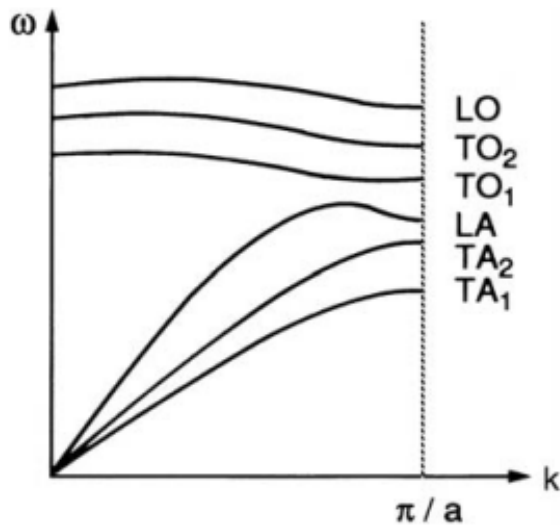


Fig. 7.15. Sketch of the dispersion relation of lattice vibrations for a three-dimensional anisotropic crystal with partly ionic binding and two atoms per primitive unit cell



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Lattice Dynamics

7.6 Quantization of Lattice Vibrations: Phonons and the Concept of Quasiparticles

- The equations with $u_n(t)$ in (7.28) or (7.23) are rather similar to the equation of motion for an harmonic oscillator.

$$m \frac{\partial^2 v_n}{\partial t^2} = -D v_n \quad (7.40)$$

- normal coordinates v_n :

$$v_n = \sum_{n'} a_{n'} u_{n'} \quad (7.41)$$

- A quantum mechanical treatment of the harmonic oscillator leads to discrete energy levels given by:

$$E_n = \hbar \omega_n \left(n + \frac{1}{2} \right), n = 0, 1, 2, 3L \quad (7.42)$$

The total energy of the lattice vibrations can be written as :

$$E = \sum_{k,i} \hbar \omega_{k,i} \left(n_{k,i} + \frac{1}{2} \right) \quad (7.43)$$

- The framework of second quantization :

$$H = \sum_{k,i} \hbar \omega_{k,i} \left(b_{k,i}^+ b_{k,i} + \frac{1}{2} \right) \quad (7.44)$$

- There is a close analogy between the quanta or particles of the electromagnetic radiation in vacuum, the photons, and the quanta of the lattice vibrations.



- The dispersion relation of phonons can be investigated by inelastic neutron scattering.

An incident neutron from a mono-energetic beam with:

$$E_i = \frac{h^2 \mathbf{k}_i^2}{2m_n} \quad (7.45)$$

Scattered, resulting in a neutron in the final state with $E_f, \mathbf{k}_f$:

$$E_f = \frac{h^2 \mathbf{k}_f^2}{2m_n} = E_i \pm \hbar \omega_{\text{phonon}}$$

$$\mathbf{k}_f = \mathbf{k}_i \pm \mathbf{k}_{\text{phonon}} + \mathbf{G} \quad (7.46)$$

Fig 7.16 shows the dispersion relation of the phonons in Si and CdS.

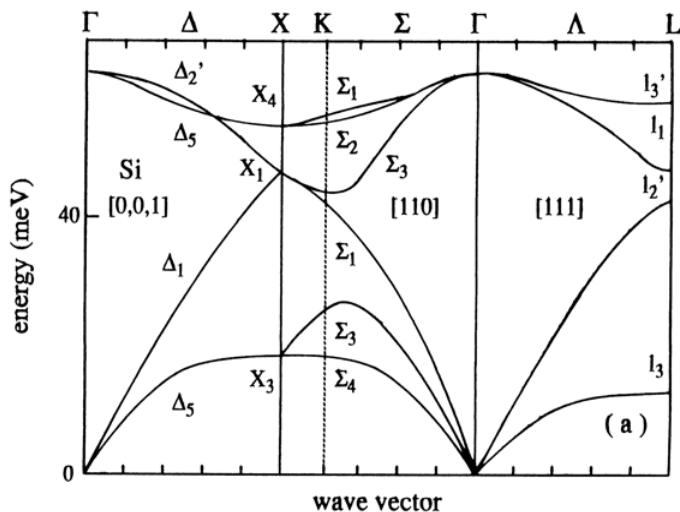
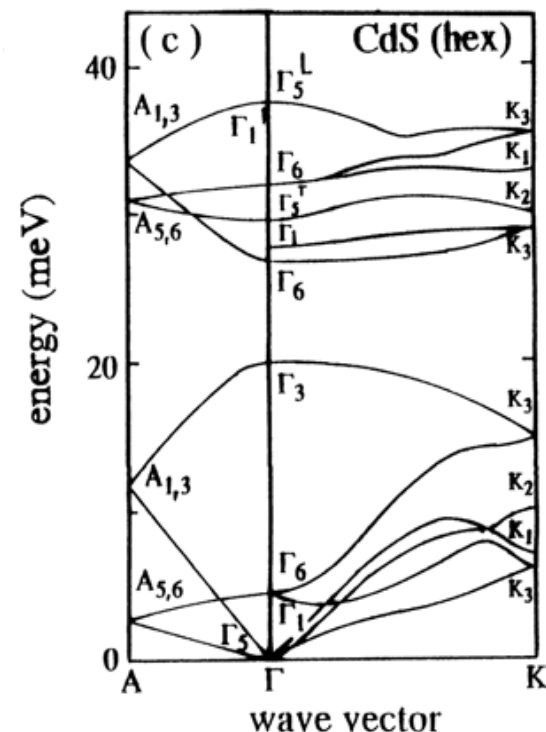


Fig. 7.16. Dispersion relation of phonons in Si and CdS for various directions in k -space



- Energy and momentum conservation in a TO phonon decays.

n or normal process:

$$(a) \hbar\omega_i = \hbar\omega_{f1} + \hbar\omega_{f2}, \mathbf{k}_i = \mathbf{k}_{f1} + \mathbf{k}_{f2} \quad \text{a TO phonon decays into two acoustic phonons} \quad (7.47a)$$

$$(b) \hbar\omega_{i1} + \hbar\omega_{i2} = \hbar\omega_f, \mathbf{k}_{i1} + \mathbf{k}_{i2} = \mathbf{k}_f \quad \text{two TA phonons combine to form a LA phonon} \quad (7.47a)$$

Umklapp or u process:

$$(c) \hbar\omega_i = \hbar\omega_{f1} + \hbar\omega_{f2}; \quad \mathbf{k}_i = \mathbf{k}_{f1} + \mathbf{k}_{f2} + \mathbf{G} \quad (7.47c)$$

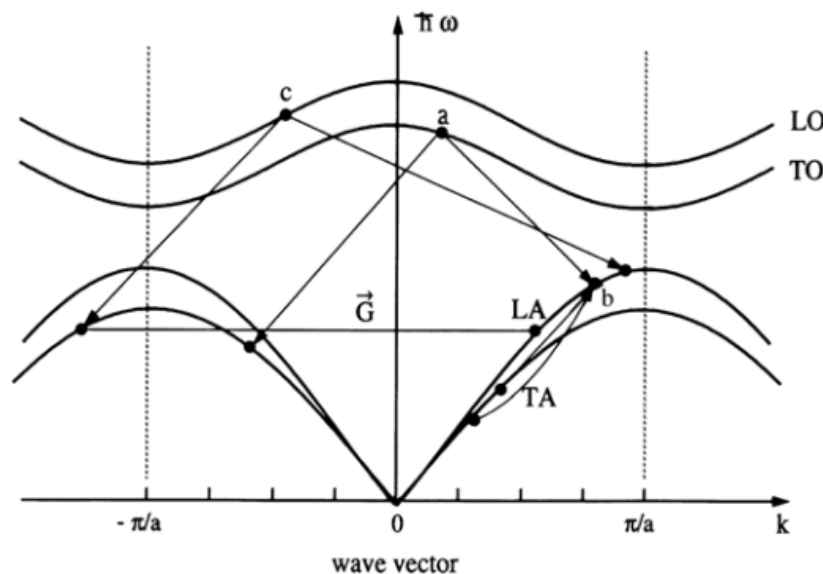


Fig. 7.17. Sketch of three possible decay or fusion processes of phonons

7.7 The Density of States and Phonon Statistics

- Now we want to calculate the density of states of phonons.

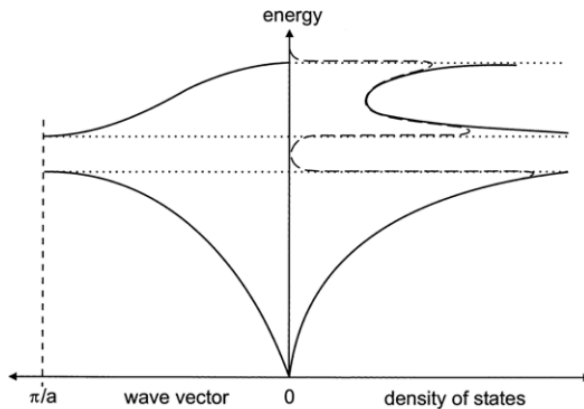


Fig. 7.18. Schematic drawing of the density of states of a three-dimensional, isotropic semiconductor with two atoms per unit cell and degenerate optical and acoustic branches

For simplicity:

1. two atoms per unit cell.
2. an isotropic semiconductor.

We need the dispersion relation to calculate $D(\omega)$.

In the linear part of the acoustic branch we have : $\omega = v_s k$ (7.48)

And it results immediately in: $D(\omega) = \text{const} \cdot \omega^2$ (7.49)

- The number of phonons :

The commutation relations for phonons obey: $[b_{k,i}, b_{k',i'}^+] = \delta_{k,k'} \delta_{i,i'}$ (7.50)

The phonons are bosons, so we have in thermal equilibrium to use Bose-Einstein statistics:

$$f_{BE}(\hbar\omega, T) = \left\{ \exp \left[\frac{(\hbar\omega - \mu)}{k_B T} \right] - 1 \right\}^{-1} \quad (7.51)$$

- In thermal equilibrium ,for the total number of phonons :

$$N(T) = \int_0^{\infty} D(\hbar\omega) \left[\exp\left(\frac{\hbar\omega}{k_B T}\right) - 1 \right]^{-1} d\omega \quad (7.52a)$$

For the energy of the phonon system:

$$U(T) = \int_0^{\infty} \hbar\omega D(\hbar\omega) \left[\exp\left(\frac{\hbar\omega}{k_B T}\right) - 1 \right]^{-1} d\omega \quad (7.52b)$$

The specific heat of the phonon system:

$$c_p \simeq c_v = \frac{\partial U}{\partial T} \quad (7.53)$$

Einstein assumed $D(\hbar\omega)$ by a δ - function:

$$D(\hbar\omega) = \delta(\omega - \omega_E) 3Ns \quad (7.54a)$$

$$3N_s = \int_0^{\omega_D} D(\hbar\omega) d\omega \quad \text{with} \quad D(\omega) \propto \omega^2 \quad (7.54b)$$

- For $k_B T > \hbar\omega_D$ or $\hbar\omega_E$: the law of Dulong and Petit:

$$c_v = 3Ns k_B \quad (7.55)$$

From the Debye's approximation one finds the well-know T^3 law:

$$c_v : (T/\theta)^3 \quad \text{with} \quad k_B \theta = \hbar\omega_D \quad \text{and} \quad T < \theta \quad (7.56)$$

- Stimulated processes for phonons:

$$W_{i \rightarrow f} : (N_{ph}^k + 1) \quad (7.57)$$

i.e., a lattice temperature of 77K, with 0.2meV , $N_{ph}^k ; 30 ? 1$

Means that: The occupation number is large.

7.8 Phonons in Alloys

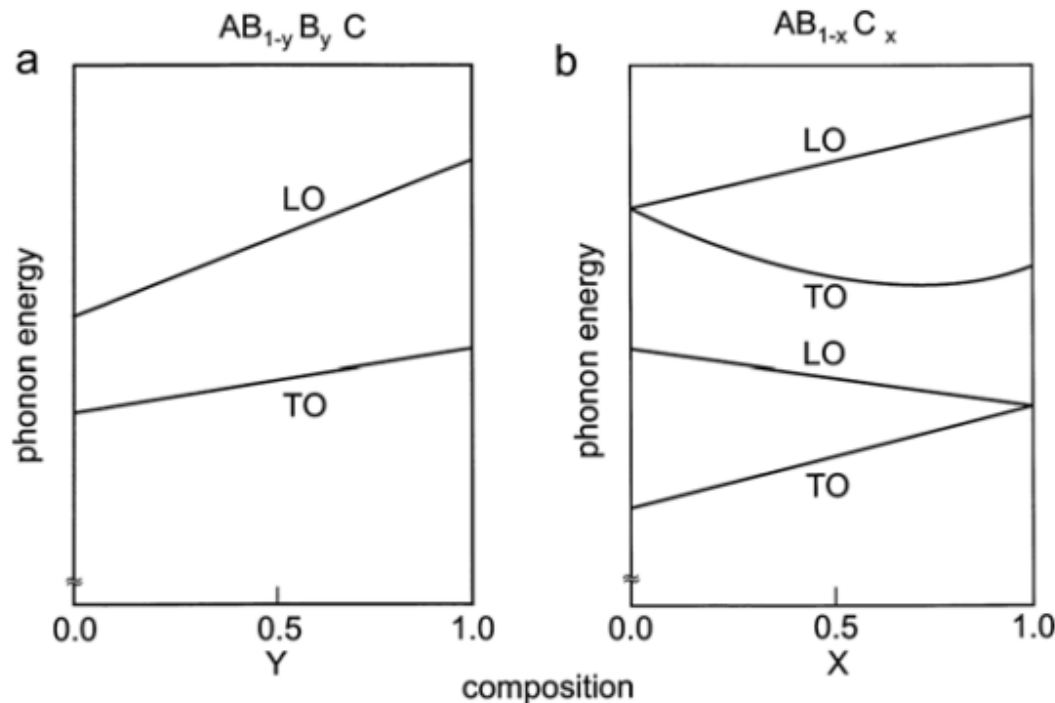


Fig. 7.19. Composition dependence of the LO phonon energies in alloys for the cases of amalgamation type (a) and persistent mode type (b) behaviour of the phonons

Acoustic phonon: velocity of sound is a weighted average of two components

Optical phonon: amalgamation mode $Zn_{1-y}Cd_yS$

persistent mode AlGaAs

附录5 元素周期

族		附录5 元素周期																0 18	电子层	0 族	
周期	I A																	2		电子数	
1	1 H 氢 $1s^1$ 1.008	II A		原子序数 — 92 U 元素符号 — 铀 注* 的是人元素 5f ⁶ 6d ¹ 7s ² 238.0 — 外围电子层排布, 括号指可能的电子层排布 — 相对原子质量(加括号的数据为该放射性元素 衰期 长同位素的质量数)										III A 13 IV A 14 V A 15 VI A 16 VII A 17					2 He 氦 $1s^2$ 4.003	K	2
2	3 Li 锂 $2s^1$ 6.941	4 Be 铍 $2s^2$ 9.012											5 B 硼 $2s^2 2p^1$ 10.81	6 C 碳 $2s^2 2p^2$ 12.01	7 N 氮 $2s^2 2p^3$ 14.01	8 O 氧 $2s^2 2p^4$ 16.00	9 F 氟 $2s^2 2p^5$ 19.00	10 Ne 氖 $2s^2 2p^6$ 20.18	L K	8 2	
3	11 Na 钠 $3s^1$ 22.99	12 Mg 镁 $3s^2$ 24.31	III B 3	IV B 4	V B 5	VI B 6	VII B 7	8	VIII 9	10	I B 11	II B 12	13 Ar 铝 $3s^2 3p^1$ 26.98	14 Si 硅 $3s^2 3p^2$ 28.09	15 P 磷 $3s^2 3p^3$ 30.97	16 S 硫 $3s^2 3p^4$ 32.06	17 Cl 氯 $3s^2 3p^5$ 35.45	18 Ar 氩 $3s^2 3p^6$ 39.95	M L K	8 8 2	
4	19 K 钾 $4s^1$ 39.10	20 Ca 钙 $4s^2$ 40.08	21 Sc 钪 $3d^1 4s^2$ 44.96	22 Ti 钛 $3d^2 4s^2$ 47.87	23 V 钒 $3d^3 4s^2$ 50.94	24 Cr 铬 $3d^5 4s^1$ 52.00	25 Mn 锰 $3d^5 4s^2$ 54.94	26 Fe 铁 $3d^6 4s^2$ 55.85	27 Co 钴 $3d^7 4s^2$ 58.93	28 Ni 镍 $3d^8 4s^2$ 58.69	29 Cu 铜 $3d^{10} 4s^1$ 63.55	30 Zn 锌 $3d^{10} 4s^2$ 65.41	31 Ga 镓 $4s^2 4p^1$ 69.72	32 Ge 锗 $4s^2 4p^2$ 72.64	33 As 砷 $4s^2 4p^3$ 74.92	34 Se 硒 $4s^2 4p^4$ 78.96	35 Br 溴 $4s^2 4p^5$ 79.90	36 Kr 氪 $4s^2 4p^6$ 83.80	N M L K	8 18 8 2	
5	37 Rb 铷 $5s^1$ 85.47	38 Sr 锶 $5s^2$ 87.62	39 Y 钇 $4d^1 5s^2$ 88.91	40 Zr 锆 $4d^2 5s^2$ 91.22	41 Nb 铌 $4d^4 5s^1$ 92.91	42 Mo 钼 $4d^5 5s^1$ 95.94	43 Tc 锝 $4d^5 5s^2$ [98]	44 Ru 钌 $4d^7 5s^1$ 101.1	45 Rh 铑 $4d^8 5s^1$ 102.9	46 Pd 钯 $4d^{10}$ 106.4	47 Ag 银 $4d^{10} 5s^1$ 107.9	48 Cd 镉 $4d^{10} 5s^2$ 112.4	49 In 铟 $5s^2 5p^1$ 114.8	50 Sn 锡 $5s^2 5p^2$ 118.7	51 Sb 锑 $5s^2 5p^3$ 121.8	52 Te 碲 $5s^2 5p^4$ 127.6	53 I 碘 $5s^2 5p^5$ 126.9	54 Xe 氙 $5s^2 5p^6$ 131.3	O N M L K	8 18 18 8 2	
6	55 Cs 铯 $6s^1$ 132.9	56 Ba 钡 $6s^2$ 137.3	57~71 La~Lu 镧系	72 Hf 铪 $5d^2 6s^2$ 178.5	73 Ta 钽 $5d^3 6s^2$ 180.9	74 W 钨 $5d^4 6s^2$ 183.8	75 Re 铼 $5d^5 6s^2$ 186.2	76 Os 锇 $5d^6 6s^2$ 190.2	77 Ir 铱 $5d^7 6s^2$ 192.2	78 Pt 铂 $5d^9 6s^1$ 195.1	79 Au 金 $5d^{10} 6s^1$ 197.0	80 Hg 汞 $5d^{10} 6s^2$ 200.5	81 Tl 铊 $6s^2 6p^1$ 204.4	82 Pb 铅 $6s^2 6p^2$ 207.2	83 Bi 铋 $6s^2 6p^3$ 209.0	84 Po 钋 $6s^2 6p^4$ (209)	85 At 砹 $6s^2 6p^5$ (210)	86 Rn 氡 $6s^2 6p^6$ (222)	P O N M L K	8 18 32 18 8 2	
7	87 Fr 钫 $7s^1$ (223)	88 Ra 镭 $7s^2$ (226)	89~103 Ac~Lr 锕系	104 Rf 𬬻 $5f^{14} 7s^2$ (261)	105 Db 𬬭 $5f^{14} 7s^2$ (262)	106 Sg 𬬮 $5f^{14} 7s^2$ (266)	107 Bh 𬬱 $5f^{14} 7s^2$ (264)	108 Hs 𬬲 $5f^{14} 7s^2$ (277)	109 Mt 𬬳 $5f^{14} 7s^2$ (268)	110 Uun Uun * (281)	111 Uuu Uuu * (272)	112 Uub Uub * (285)	注：相对原子质量录自2001年国际原子量，并全部取4位有效数字。								

镧系	57 La 镧 $5d^1 6s^2$ 138.9	58 Ce 铈 $4f^1 5d^1 6s^2$ 140.1	59 Pr 镨 $4f^3 6s^2$ 140.9	60 Nd 钕 $4f^4 6s^2$ 144.2	61 Pm 钷 $4f^5 6s^2$ (145)	62 Sm 钐 $4f^6 6s^2$ 150.4	63 Eu 铕 $4f^7 6s^2$ 152.0	64 Gd 钆 $4f^7 5d^1 6s^2$ 157.3	65 Tb 铽 $4f^9 6s^2$ 158.9	66 Dy 镝 $4f^{10} 6s^2$ 162.5	67 Ho 铒 $4f^{11} 6s^2$ 164.9	68 Er 铈 $4f^{12} 6s^2$ 167.3	69 Tm 铥 $4f^{13} 6s^2$ 168.9	70 Yb 镱 $4f^{14} 6s^2$ 173.0	71 Lu 镥 $4f^{14} 5d^1 6s^2$ 175.0
锕系	89 Ac 锕 $6d^1 7s^2$ (227)	90 Th 钍 $6d^2 7s^2$ 232.0	91 Pa 镤 $5f^2 6d^1 7s^2$ 231.0	92 U 铀 $5f^3 6d^1 7s^2$ 238.0	93 Np 镎 $5f^4 6d^1 7s^2$ (237)	94 Pu 钚 $5f^6 7s^2$ (244)	95 Am 镅 $5f^7 7s^2$ (243)	96 Cm 锔 $5f^7 6d^1 7s^2$ (247)	97 Bk 锫 $5f^9 7s^2$ (247)	98 Cf 锿 $5f^{10} 7s^2$ (251)	99 Es 镅 $5f^{11} 7s^2$ (252)	100 Fm 镆 $5f^{12} 7s^2$ (257)	101 Md 钔 $5f^{13} 7s^2$ (258)	102 No 镎 $5f^{14} 7s^2$ (259)	103 Lr 铹 $5f^{14} 6d^1 7s^2$ (262)



7.9 Defects and Localized Phonon Modes

- In fact , every real semiconductor contains a lot of crystal or lattice defects.

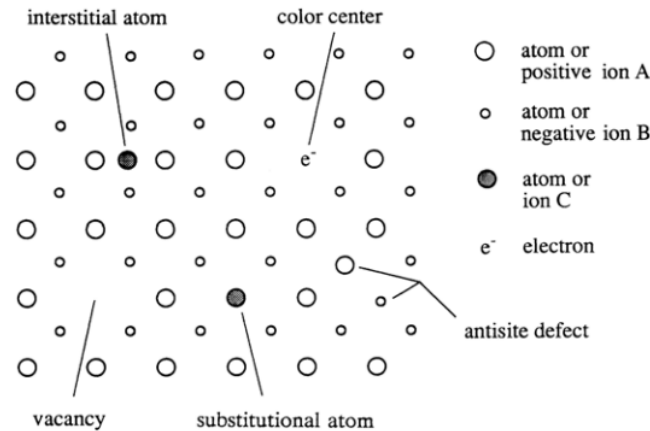


Fig. 7.20. The lattice of a semiconductor AB containing various types of point defects

- A consequence : a localized phonon mode may appear. $\omega_{\text{loc}} > \omega_{\text{bulk}}^{\text{LO}}$ (7.59)
- A localized mode can be constructed in a Fourier-transform-like method:

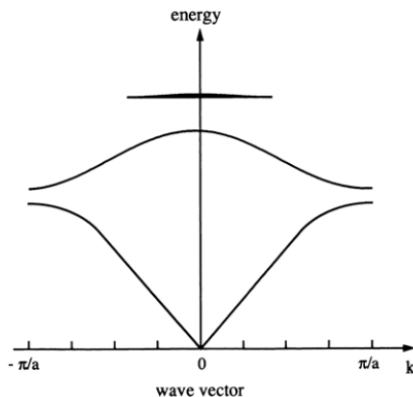


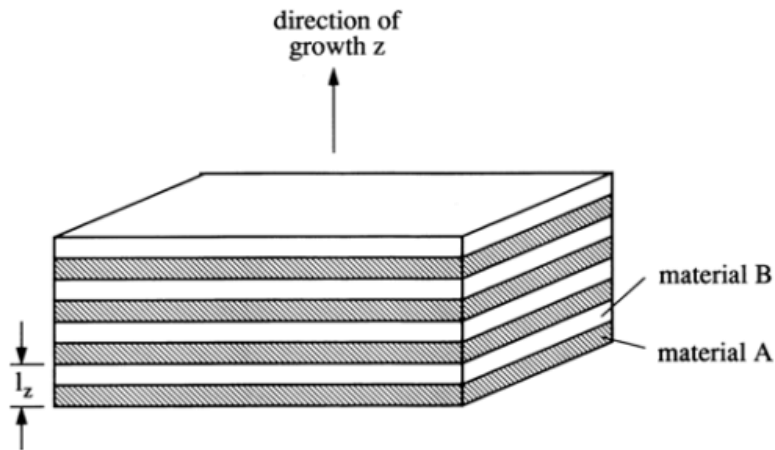
Fig. 7.21. A localized phonon mode represented schematically in a dispersion diagram

$$u_{\text{loc}}(\mathbf{r}) = \sum_{\mathbf{k}, i} a_{\mathbf{k}, i} u_{\mathbf{k}, i}(\mathbf{r}) \quad (7.60)$$

Every defect is a scattering center for phonons and contributes, together with the anharmonicity mentioned above, to the finite phase relaxation time of phonons.

7.10 Phonons in Superlattices and in other Structures of Reduced Dimensionality

A super-lattice is a man-made periodic structure which consists of thin alternating layers of two different materials.



Materials:

1. the III–V compounds GaAs, AlAs and their alloys $\text{Al}_{1-y}\text{Ga}_y\text{As}$
2. Si and Ge or ZnSe and $\text{ZnS}_{1-x}\text{Se}_x$.

Fig. 7.22. A superlattice consisting of alternating thin layers of two different materials

From Fig.7.22, a new artificial periodicity: $l_z = n_1 a_A + n_2 a_B$ (7.62)

The Brillouin zone in the z-direction: $-\frac{\pi}{l_z} \leq k_z \leq \frac{\pi}{l_z}$ (7.63a)

Vectors of the new reciprocal lattice:

$$G_z = l_z \frac{2\pi}{l_z}; l_z = 0, \pm 1, \pm 2, \dots \quad (7.63b)$$

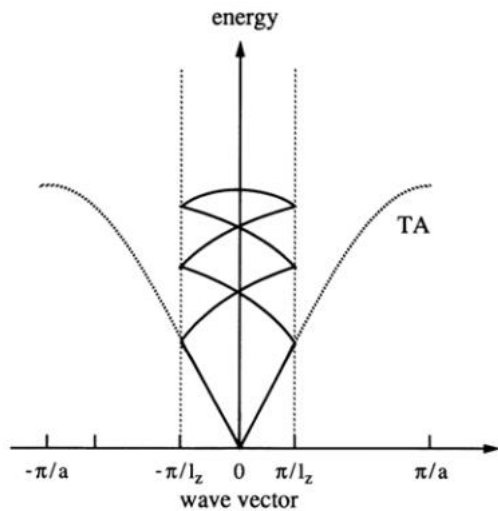


Fig. 7.23. The influence of superlattice formation on the acoustic phonon branch

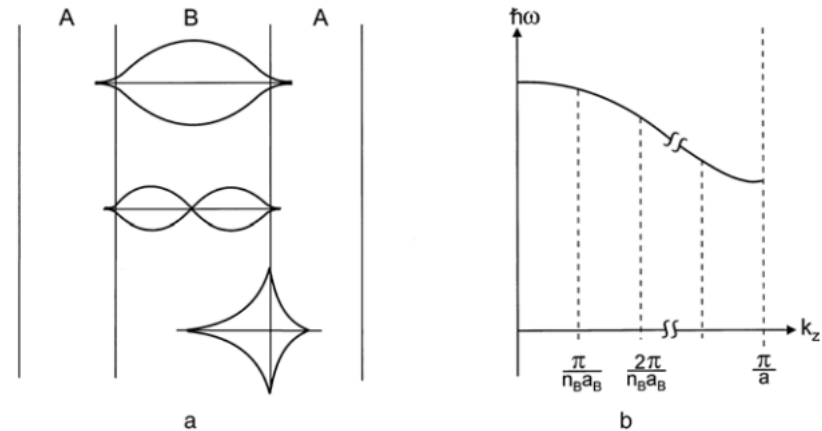
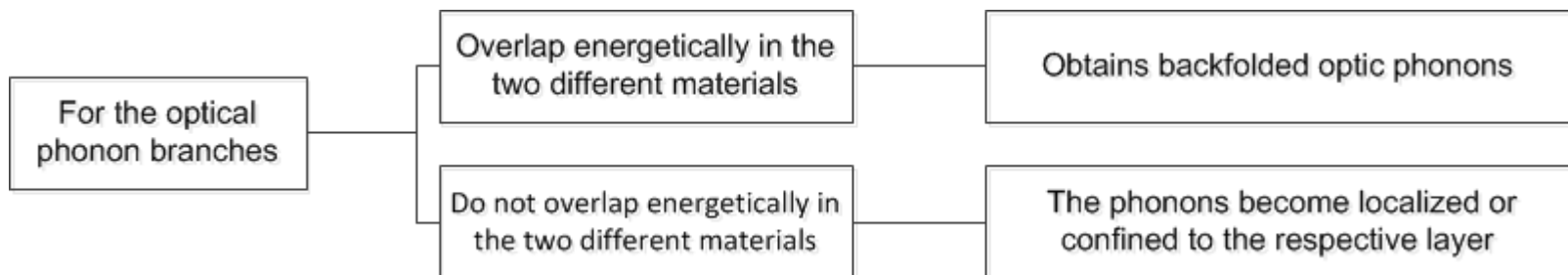


Fig. 7.24. Schematic representation of the envelope of an optical phonon mode confined to material B in a superlattice and of an interface phonon (a) and the discrete values of k_z on an optical phonon branch of material B. (b)

- Fig.7.23 for a single acoustic phonon branch
- For optical phonon branches



In Fig.7.24 , possible values: $k_z^{(m)} = m \frac{\pi}{n_i a_i}; m = 1, 2, 3K \quad i = A, B$ (7.64)

- In superlattices there may be as new phenomena backfolded or confined phonon modes as well as interface modes.